

Changes in Physical Performance Throughout Professional Soccer Match-Play

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Abstract

Oliva-Lozano, JM, Granero-Gil, P, and Panasci, M. Changes in physical performance throughout professional soccer match-play. *J Strength Cond Res* XX(X): 000–000, 2023—The aim of this study was to explore the changes in physical performance by playing position throughout professional soccer match-play. Global Positioning Systems were used for the collection of external load measures during 35 matches. Each match was divided into 15-minute periods. Physical performance varied according to the period in distance covered ($F = 25.69$; $p < 0.001$; partial eta-squared [η^2] = 0.11), sprinting distance (SPD, $F = 4.35$; $p < 0.001$; $\eta^2 = 0.02$), sprinting actions (SPAs, $F = 5.19$; $p < 0.001$), high-intensity accelerations (Acc+3, $F = 2.18$; $p = 0.05$; $\eta^2 = 0.01$), high-intensity decelerations (Dec+3, $F = 4.51$; $p < 0.001$; $\eta^2 = 0.02$), and high-metabolic load distance (HMLD, $F = 5.59$; $p < 0.001$; $\eta^2 = 0.03$). The period with the greatest mean physical performance was usually from the first half of the matches: 0–15 minutes (distance, Acc+3, and Dec+3) and 30–45 minutes (HMLD, SPA, and SPD). However, the lowest mean physical performance compared with the rest of 15-minute periods was observed in the 60–75-minute period (distance, HMLD, Acc+3, and Dec+3), except for SPA and SPD (15–30 minutes and 75–90 minutes). Overall, no significant differences ($p > 0.05$) were observed in the physical performance between the extratime periods. In addition, the playing position had a significant effect on all performance variables ($F = 39.99$ – 133.29 ; $p < 0.001$; $\eta^2 = 0.14$ – 0.34). Central defenders showed the lowest physical performance throughout the periods of match-play, whereas wide midfielders showed the greatest physical performance in all variables, except for distance covered, in which midfielders had the greatest performance. In conclusion, there were changes in physical performance during the matches depending on the period of match-play and playing position.

Key Words: GPS, game, evolution, fatigue, monitoring, period

Introduction

The use of Global Positioning Systems (GPS) in soccer teams allows to measure any variable related to relationship between time and positioning such as distance covered, velocity, or acceleration during the training sessions and official matches of each player (3). This technology allows to keep an accurate balance between training stress and recovery, thereby maximizing performance potential while minimizing the risk of overtraining (12). This is why the interest in the application of technology for monitoring the physical performances has been significantly developed in recent years.

Soccer is characterized by periods of high-intensity activity that are interspersed with lower intensity actions (23). During a match, players perform different activities such as changing of direction, jumping, shooting, tackling, sprinting, controlling the ball under pressure, and running at different speeds, which involve high metabolic (aerobic and anaerobic energy production) and physiological requirements (10). Soccer players who complete the full duration of the match may cover ~10.2 km and ~600 m at high-intensity running (14). Specifically, sprinting actions (SPAs) usually have a tactical purpose, which is specific to each playing position (20), and may lead to situations like scoring a goal (8).

Previous research observed that the capacity to perform accelerations and decelerations, or sprints decreases during the last 15 minutes of match-play (22,27). The level of glycogen in players' muscle fibers decreases during the match causing a reduction of the number of sprints, accelerations, and decelerations (13,28). In addition, several studies demonstrated that physical performance during competitive matches is influenced by the playing position (4–6). For example, the total distance covered and high-intensity distance covered by midfielders (MFs), full-backs (FBs), and wide midfielders (WMFs) during a soccer match are higher than the distances covered by forwards (FWs) or central defenders (CDs) (6). In addition, the highest sprinting distances (SPDs) through a competitive soccer match are achieved by the players disposed in wide position on the pitch compared with the central players (11).

Physical performance of match day has been examined using different external load parameters, considering contextual variables such as playing position (1,4) or players' level (17), and dividing the full match into different periods (7,21). However, to the best of our knowledge, little research is available analyzing a combination of external load parameters with a specific focus on high-intensity actions such as SPD, SPAs, high-intensity accelerations (Acc+3), high-intensity decelerations (Dec+3), and high-metabolic load distance (HMLD), dividing the duration of the full match into 15-minute periods. This information would allow coaches to design specific training strategies that allow the players to maintain a good level of performance across various phases of the game.

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Therefore, the aims of this study were (a) to analyze the changes in physical performance of the professional soccer players during the different 15-minute periods of match and (b) to investigate the effect that the playing position had on the changes in physical performance throughout match-play. The hypotheses were that the greatest performance would be during the first 15-minute period of the match regardless of the playing position and that the lowest physical performance would be found toward the end of the match (60–75 minutes and 75–90 minutes).

Methods

Experimental Approach to the Problem

This was an observational longitudinal study that considered 35 matches of a professional soccer team competing in the first league from Hungary. Performance tracking technology, which was based on the GPS, was used for data collection during match days. To explore the changes in physical performance throughout match-play, each match was divided into periods of 15 minutes. The extratime from each match was also included as an additional period for each match half, although the length of this period varied according to the matches. Before the beginning of the experimental protocol, all subjects were fully informed about the study aims and procedure, and they provided their written informed consent to participate in the study. The club authorized the data collection during the season and the experimental protocol conformed to the code of Ethics of the World Medical Association (Declaration of Helsinki III), and it was approved by the Ethics Committee of the University of Almeria (UALBIO2020/032).

Subjects

Twenty-six professional soccer players (age: 27.3 ± 2.7 years; height: 1.8 ± 0.1 m; body mass: 78.8 ± 6.6 kg) took part in the study. The following playing positions were included in the study: MFs ($n = 8$), CDs ($n = 6$), FBs ($n = 4$), FWs ($n = 4$), and WMFs ($n = 4$). The players were excluded from the analysis when the total duration of the match was not completed.

Procedures

WIMU Pro units (RealTrack Systems, Almería, Spain) were used for data collection of the matches. These units were based on GPS technology that collected data at 10 Hz. All units were calibrated before the start of the session following the manufacturer's instructions (19,24). Then, 1 member of the research team placed the tracking systems in the back pocket of a chest vest (Rasán, Valencia, Spain). The same member of the research team transferred the data to SPro software (RealTrack Systems) through the Smart Station (RealTrack Systems) after each match. The SPro software created the "Intervals Pro" report and the following variables were analyzed: total distance covered (m), HMLD (meters covered when player's metabolic power was above $25.5 \text{ W} \cdot \text{kg}^{-1}$), Acc+3 (above $3 \text{ m} \cdot \text{s}^{-2}$), Dec+3 (below $-3 \text{ m} \cdot \text{s}^{-2}$), SPD (meters covered above $25.2 \text{ km} \cdot \text{h}^{-1}$), and SPAs (above $25.2 \text{ km} \cdot \text{h}^{-1}$) (23).

Statistical Analyses

The normality of the variables was assessed through Kolmogorov-Smirnov and homogeneity of variance through the Levene's test. Descriptive statistics (mean and SD) were calculated to obtain the physical performance required throughout professional soccer matches. A 2-way analysis of variance with the

Bonferroni post hoc test was conducted to analyze the changes in physical performance during the match. Playing position and the periods of the match (i.e., each 15-minute period for 90 minutes) were set as independent variables. The extratime periods for each match half were removed from this analysis because these periods are shorter, do not have a regular duration, and could likely affect the outcomes of the 2-way analysis. Partial eta-squared (η^2) was used to analyze the effect size (large effect: ≥ 0.14 , medium effect: $0.06\text{--}0.14$, and small effect: $0.01\text{--}0.06$). The statistical analysis was conducted on SPSS Statistics version 28 (IBM Corp., Armonk, NY), and the statistical significance was set at $p < 0.05$.

Results

Figure 1 shows the changes in physical performance throughout professional soccer match-play. The physical performance varied according to the period of the match in all variables: distance covered ($F = 25.69$; $p < 0.001$; $\eta^2 = 0.11$, medium effect), SPD ($F = 4.35$; $p < 0.001$; $\eta^2 = 0.02$, small effect), SPAs ($F = 5.19$; $p < 0.001$; $\eta^2 = 0.03$, small effect), Acc+3 ($F = 2.18$; $p = 0.05$; $\eta^2 = 0.01$, medium effect), Dec+3 ($F = 4.51$; $p < 0.001$; $\eta^2 = 0.02$, small effect), and HMLD ($F = 5.59$; $p < 0.001$; $\eta^2 = 0.03$, small effect). Specifically, the period with the greatest mean physical performance was usually from the first half of the matches: 0–15 minutes (distance: 1801.29 ± 183.10 m; Acc+3: 9.05 ± 4.01 accelerations; Dec+3: 12.56 ± 5.06 decelerations) and 30–45 minutes (HMLD: 172.32 ± 63.86 m; SPAs: 1.69 ± 1.42 actions; SPD: 33.53 ± 32.04 m). However, the lowest mean physical performance compared with the rest of 15-minute periods was observed in the 60–75-minute period (distance: $1,546.97 \pm 195.81$ m; HMLD: 145.41 ± 54.45 m; Acc+3: 7.68 ± 3.20 accelerations; Dec+3: 10.21 ± 4.16 decelerations), except for SPA (15–30 minutes: 1.20 ± 1.29 actions; 75–90 minutes: 1.21 ± 1.29 actions) and SPD (15–30 minutes: 23.99 ± 27.46 m; 75–90 minutes: 24.51 ± 28.63 m).

The playing position had a significant effect on all variables: distance covered ($F = 83.87$; $p < 0.001$; $\eta^2 = 0.25$, large effect), SPD ($F = 64.58$; $p < 0.001$; $\eta^2 = 0.20$, large effect), SPA ($F = 69.72$; $p < 0.001$; $\eta^2 = 0.21$, large effect), ACC_{HIGH} ($F = 39.99$; $p < 0.001$; $\eta^2 = 0.14$, large effect), DEC_{HIGH} ($F = 78.63$; $p < 0.001$; $\eta^2 = 0.24$, large effect), and HMLD ($F = 133.29$; $p < 0.001$; $\eta^2 = 0.34$, large effect). Specifically, CDs showed the lowest physical performance throughout the periods of match-play, whereas WMFs showed the greatest physical performance in all variables, except for distance covered, in which MFs had the greatest performance. Table 1 shows the changes in physical performance throughout professional soccer match-play by playing position. However, no significant interactions between playing position and the periods of the match were observed for any of the variables: distance covered ($F = 0.36$; $p = 0.99$; $\eta^2 = 0.01$, small effect), SPD ($F = 1.53$; $p = 0.06$; $\eta^2 = 0.03$, small effect), SPA ($F = 1.55$; $p = 0.06$; $\eta^2 = 0.03$, small effect), ACC_{HIGH} ($F = 0.62$; $p = 0.90$; $\eta^2 = 0.01$, small effect), DEC_{HIGH} ($F = 1.42$; $p = 0.11$; $\eta^2 = 0.03$, small effect), or HMLD ($F = 0.40$; $p = 0.99$; $\eta^2 = 0.01$, small effect).

Discussion

The purpose of this study was to investigate the change in physical performance throughout professional soccer match-play. This investigation shows that there were changes in physical performance during the matches depending on the period. Overall, the period with the greatest mean physical

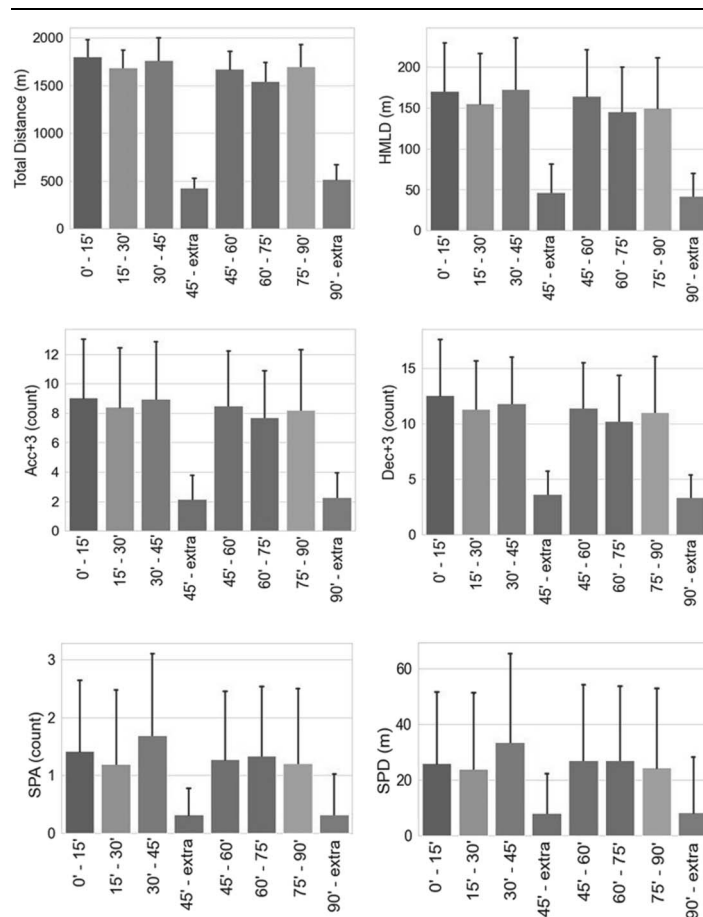


Figure 1. Description of physical performance throughout professional soccer match-play. m, total distance; HMLD, high-metabolic load distance; Acc+3, high-intensity accelerations; Dec+3, high-intensity decelerations; SPD, sprinting distance; SPAs, sprinting actions; CDs, central defenders; FBs, full-backs; FWs, forwards; MFs, midfielders; WMFs, wide midfielders.

performance was usually from the first half of the matches: 0–15 minutes (distance, Acc+3, and Dec+3) and 30–45 minutes (HMLD, SPA, and SPD). However, the lowest mean physical performance compared with the rest of the 15-minute periods was observed in the 60–75-minute period (distance, HMLD, Acc+3, and Dec+3), except for SPA and SPD (15–30 minutes and 75–90 minutes). The playing position had a significant effect on all physical performance variables. Central defenders showed the lowest physical performance throughout the periods of match-play, whereas WMFs showed the greatest physical performance in all variables, except for distance covered, in which MFs had the greatest performance. However, the interaction between playing position and period of the match was not significant for any of the variables.

The results of our study demonstrated that the highest values of distance, Acc+3, and Dec+3 have been reached during the first period of the match (0–15 minutes). These parameters are probably influenced by factors such as the desire to obtain a tactical superiority on the opponent team, positive ergogenic effects of the warm-up prematch, or the higher level of neuromuscular freshness in the first 15 minutes of the match. Moreover, our data agree with those obtained in a previous study (27), which observed that the distance covered and the high-intensity acceleration and deceleration distances performed by the soccer players were higher during the first 15 minutes of the

match than those reached in the other bouts of the match. Furthermore, the present data, in average, indicated that HMLD, SPD, and SPA reached during the third period (30–45 minutes) were significantly higher than those measured during the other periods of the match. These variables usually go together because the sprinting activity is related to the HMLD (although, to the best of our knowledge, there are no other studies evaluating the differences of this variable during the different periods of the match), which represents the meters covered when player's metabolic power is high (e.g., above $25.5 \text{ W} \cdot \text{kg}^{-1}$) (18). Concerning this, previous studies conversely have indicated that the greatest sprint performance was found during the first (0–15 minutes) and fourth (45–60 minutes) periods, with a reduction in performance during the last phases of each match half (17,22). For example, Mohr et al. (17) found that in top-level Danish professional soccer players, the SPD performed ($30 \text{ km} \cdot \text{h}^{-1}$) during the match was higher during the first period of the match than the other periods. In addition, a recent study showed that the greatest amount of sprints above $30 \text{ km} \cdot \text{h}^{-1}$ in Spanish LaLiga were found in 0–15 minutes and 75–90 minutes (20). The difference of results between previous research and our data could be explained by different reasons, such as effect of contextual variables, the use of optical tracking methods vs. data collected by GPS, or the different categorization of sprint thresholds.

Table 1**Evolution of physical performance during professional soccer match-play by playing position.***

Variable	Position	0'-15'	15'-30'	30'-45'	45'-60'	60'-75'	75'-90'
Total distance (m)	CDs	1,667.49 ± 125.74†‡§	1,561.50 ± 154.46§¶	1,639.74 ± 232.90§	1,546.80 ± 155.76§¶	1,423.34 ± 175.41†‡ ¶#	1,552.13 ± 200.36§¶
	FBs	1856.73 ± 180.78†‡§	1727.61 ± 186.08§¶	1803.93 ± 208.02§	1,692.32 ± 174.79¶	1,577.08 ± 217.07†‡ ¶#	1762.77 ± 199.66§
	FWs	1732.04 ± 104.11	1,616.03 ± 170.39	1724.94 ± 200.44	1,666.16 ± 126.54	1,564.44 ± 101.59	1,635.72 ± 240.90
	MFs	1942.85 ± 129.36†‡§	1817.70 ± 160.76§¶	1885.58 ± 214.62§	1799.25 ± 147.72§¶	1,661.37 ± 149.56†‡ ¶#	1835.06 ± 191.13§
	WMFs	1734.03 ± 171.26§	1,628.73 ± 154.81	1752.77 ± 247.52§	1,660.04 ± 186.64	1,527.66 ± 154.67¶#	1,648.67 ± 260.59
SPD (m)	CDs	19.40 ± 17.38	9.86 ± 16.92	14.72 ± 18.83	10.45 ± 14.11	13.17 ± 18.87	8.70 ± 14.73
	FBs	35.49 ± 23.73	29.65 ± 28.76#	50.19 ± 30.39†	39.21 ± 31.31	37.00 ± 30.64	32.79 ± 28.69#
	FWs	22.03 ± 17.16#	35.70 ± 37.55	60.16 ± 41.98¶	44.78 ± 29.76	37.74 ± 23.09	41.17 ± 32.10
	MFs	21.88 ± 24.73	21.29 ± 20.35	25.45 ± 26.12	24.75 ± 25.00	20.00 ± 19.06	21.00 ± 24.19
	WMFs	37.24 ± 43.57	50.41 ± 33.26	54.48 ± 34.10	41.03 ± 27.93	54.09 ± 26.34	50.62 ± 38.08
SPAs (count)	CDs	1.00 ± 0.82	0.44 ± 0.69	0.81 ± 0.93	0.54 ± 0.67	0.67 ± 0.78	0.48 ± 0.77
	FBs	1.95 ± 1.23	1.43 ± 1.27#	2.43 ± 1.43†	1.81 ± 1.35	1.76 ± 1.32	1.69 ± 1.41#
	FWs	1.73 ± 1.19	1.73 ± 1.85	3.00 ± 1.61	1.73 ± 1.01	2.09 ± 1.38	1.64 ± 1.21
	MFs	1.31 ± 1.36	1.16 ± 1.09	1.35 ± 1.01	1.16 ± 1.11	1.06 ± 0.92	1.10 ± 1.20
	WMFs	1.58 ± 1.47#	2.55 ± 1.40	2.65 ± 1.46¶	2.15 ± 1.04	2.55 ± 1.05	2.26 ± 1.37
Acc+3 (count)	CDs	7.00 ± 2.66	6.56 ± 3.48	7.54 ± 4.03	6.83 ± 3.13	5.93 ± 2.49	6.59 ± 3.40
	FBs	9.93 ± 4.30	8.45 ± 4.07	8.90 ± 3.18	8.48 ± 3.79	8.24 ± 3.43	8.62 ± 4.71
	FWs	9.27 ± 4.69	8.55 ± 3.14	8.09 ± 4.25	7.55 ± 3.45	8.64 ± 3.20	6.73 ± 3.23
	MFs	9.84 ± 4.24	8.90 ± 3.72	9.35 ± 3.77	9.16 ± 3.25	8.06 ± 3.04	8.80 ± 3.42
	WMFs	10.74 ± 3.65	12.10 ± 3.91	12.45 ± 2.95	11.95 ± 4.10	9.80 ± 2.84	11.16 ± 4.73
Dec+3 (count)	CDs	8.96 ± 3.07	8.70 ± 3.35	9.46 ± 3.46	9.54 ± 3.40	8.06 ± 3.15	8.61 ± 3.48
	FBs	14.31 ± 4.58†§	11.55 ± 3.84¶	12.12 ± 3.04	12.64 ± 3.64	11.31 ± 4.52¶	12.60 ± 4.92
	FWs	9.82 ± 2.40	10.55 ± 3.17	11.45 ± 3.73	9.36 ± 3.30	9.91 ± 4.21	8.73 ± 4.67
	MFs	13.96 ± 5.18†§	12.29 ± 4.03	12.20 ± 4.08	10.86 ± 3.71¶	10.43 ± 3.74¶	10.71 ± 4.37¶
	WMFs	16.89 ± 4.59	15.85 ± 4.68	16.70 ± 4.51	16.20 ± 3.65	13.35 ± 4.04	16.58 ± 5.51
HMLD (m)	CDs	125.40 ± 34.54	105.67 ± 35.19	123.13 ± 43.95	118.94 ± 38.70	104.38 ± 39.40	102.26 ± 35.41
	FBs	199.83 ± 60.72§	176.68 ± 58.01	192.37 ± 52.03	186.07 ± 55.39	165.15 ± 53.01¶	164.95 ± 49.99¶
	FWs	165.97 ± 46.58	165.59 ± 59.15	186.08 ± 62.53	165.37 ± 32.56	158.05 ± 41.77	159.62 ± 74.48
	MFs	176.60 ± 56.98§	163.28 ± 52.23	178.97 ± 56.46§	169.00 ± 43.07	147.33 ± 42.64¶#	159.34 ± 55.56
	WMFs	216.69 ± 52.25	220.83 ± 47.69	239.17 ± 59.46	230.12 ± 52.57	204.02 ± 45.15	220.15 ± 57.84

*m, total distance; HMLD, high-metabolic load distance; Acc+3, high-intensity accelerations; Dec+3, high-intensity decelerations; SPD, sprinting distance; SPAs, sprinting actions; CDs, central defenders; FBs, full-backs; FWs, forwards; MFs, midfielders; WMFs, wide midfielders.

†15'-30'.

‡45'-60'.

§60'-75'.

||75'-90'.

¶Statistical differences ($p < 0.05$) from period 1 (0'-15').

#30'-45'.

Furthermore, this study showed that the lowest mean physical performance compared with the rest of the 15-minute periods was generally observed in the second half. Previous studies suggested that fatigue decreased the capacity to maintain the same physical work throughout the soccer match (7,13,17). In particular, during the last phase of the match, the neuromuscular fatigue significantly increases, which induces a reduction of physical performance of the players (9). For example, the players reached the lowest amount of Acc+3 and Dec+3 in the 60-75-minute period. A reduction of acceleration and deceleration ability through competitive match is induced by the decrease of glycogen level, lower concentration of muscle creatine-phosphate, and dehydration provoking physical fatigue, which induces a decline in the capacity to develop the muscular force (2). Our study found the lowest values in the 60-75-minute period for all variables, except for SPA and SPD (15-30 minutes and 75-90 minutes). From our perspective, fatigue should not be considered as the only 1 variable explaining this decline in performance. The effective playing time is a key variable because it is reduced in the last phases of the match (e.g., 0-15 minutes: 66.3% of the total playing time; 75-90 minutes: 55% of the total playing time) (15).

Another main finding of the study was that the playing position had a significant effect on all performance variables,

with the CDs and WMFs showing the lowest and greatest, respectively, physical performance throughout the periods of match-play, except for distance covered, in which MFs had the greatest performance. One of main reasons that may explain these results is tactical purpose of each position and density of players in the area in which each player is usually located (16,19,20). For example, these results agree with some previous studies showing that external playing positions (e.g., FBs or WMFs) may achieve greater high-intensity actions than internal positions (e.g., CDs or MFs) (4,19,20,26,28). Nonetheless, a recent study suggested that there were differences in physical performance depending on team formation (16). For instance, formations with 3 defenders increased the SPD of CDs and FBs compared with formations with 4 defenders (16).

The present study has some limitations. First, GPS technology was used for the data collection. Second, the sprint analysis is limited to a specific speed, acceleration, and deceleration threshold. Third, this study did not measure physiological measures (e.g., heart rate response) of the players during the match, which may be relevant parameters of performance (29). Fourth, the effect of the score during the 15-minute periods of the matches on performance was not analyzed. Finally, to expand the analysis of running

performance in different periods of the match, it could be interesting to conduct a similar study considering the effective playing time (25).

Practical Applications

The study shows that there were changes in physical performance during the matches depending on the period of match-play. Coaches should pay attention to neuromuscular activation during warm-ups and consider the time difference until the start of the match because the period with the greatest mean physical performance was usually 0–15 minutes for distance, Acc+3, and Dec+3. This study also found that the lowest mean physical performance compared with the rest of the 15-minute periods was observed in the 60–75-minute period (distance, HMLD, Acc+3, and Dec+3), except for SPA and SPD (15–30 minutes and 75–90 minutes). The data collection of the players' physical performance may also increase the capacity to prevent the muscle injuries, but these results should be interpreted with caution by the soccer fitness coaches because the decline of physical performance during the match may be due to game interruptions and lower effective playing time. In addition, this investigation observed that playing position had a significant effect on all physical performance variables, which suggests that an individualized training approach is necessary to optimize performance and reduce injury risk. For instance, a CD usually does not sprint as much as a WMF, and the tactical purpose and context in which they develop the actions are different.

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